Assessment of Administration of Aqueous Extract of *Garcinia Kola* on the Liver and Kidney Function Parameters in Sprague Dawley Rats

[*1]Adisa W.A., [2]Blackie O.H., [3]Idowu A.A.

Department of [1]Physiology, [2]Anatomy, Faculty of Basic Medical Sciences, College of Medicine, Ambrose Alli University, Ekpoma, Edo State, Nigeria.
[3]Department of Physiology, Faculty of Basic Medical Sciences, College of Medicine, University of Ibadan, Ibadan, Nigeria.

ABSTRACT

This study investigated the effect of aqueous seed extract of *Garcinia kola* on the liver and kidney functions in rats. Eighteen female adults rats were divided into 3 groups; a control, and two test groups treated with 0.5 or 1 mg/kg *Garcinia kola* extract using orogastric canula. Two weeks after treatment, blood samples were collected to estimate liver and kidney functions using standard laboratory procedures and the data analysed for significant different at $p<0.05$. Comparatively, there was a significant increase in serum alanine transaminase (ALT; 18.60±1.96 vs 16.00±1.79 vs 26.00±1.41 u/L), potassium (4.16±0.25 vs 4.62±0.11 vs 5.58±0.32 mmol/L), urea (27.60±12.34 vs 41.20±13.67 vs 110.00±14.65 g/dL) and creatinine (0.72±0.04 vs 0.88±0.37 vs 1.76±0.05g/dL) but no significant increase in aspartate transaminase (AST), Bilirubin, Albumin, Total Protein, sodium, chloride and bicarbonate compared with control. *Garcinia kola* extract has stimulating effect on serum ALT, urea and creatinine at a high dose and therefore should be consumed with moderation.

Keywords: *Garcinia kola,* Bilirubin, Sodium, Bicarbonate, Urea and creatinine.

***Correspondence: Tel.:** +2348056049158**; E-mail:** *williamsadewumi@yahoo.com*

1. INTRODUCTION

Garcinia kola is a fruit-bearing plant belonging to the family Guttiferae. *Garcinia kola* seed is called "bitter kola" in Nigeria due to its bitter taste. It is of high value in African Ethnomedicine due to the fact that it is used not only as a cure for different ailments in human but also for preventing diseases. It has been reported that *Garcinia kola* seed possesses anti-inflammatory, antioxidant, antidiabetic, antihepatotoxic, purgative, antiparasitic, antimicrobial properties, antithrombotic and antibacterial effect [1 - 4]. Studies have also shown its usefulness in the treatment of abdominal pain, cough, laryngitis, infectious diseases, erectile problems, lipid disorders, dysentery, diarrhoea, liver disorders, hypertention, bronchitis, throat infections, colic, head or chest colds, and cough [5, 6]. There are also reports that *Garcinia kola*, alleviates type 2 diabetes mellitus combined with nonalcoholic fatty liver disease [7], reduce *glutathione* concentration, inhibits prostaglandin synthesis [8], relaxes the smooth muscles of the uterus and gastrointestinal tract [9], stimulates histamine dependent gastric acid secretion [10], stabilizes the membranes of HbAA, HbAS and HbSS human erythrocytes and reduce blood viscosity [11]. It has also been shown to reduce body weight, reproductive organ weight and inhibit spermatogenesis in male Wistar rats [12].

Chronic ingestion of *Garcinia kola* seed was observed to induce histopathological changes in liver parenchymal cells, renal tubular epthelium and duodenal villous epithelium [13, 14]. Phytochemically *Garcinia kola* seed has been reported to contain a variety of phytochemicals including flavonoids, saponins, tannins and cardiac glycosides [15]. The flavonoid of *Garcinia kola* seed, also known as *kolaviron* has been identified as its

main bioactive component which is responsible for the numerous physiological and pharmacological effects.

In view of the medicinal effects of *Garcinia kola,* and bearing in mind its abuse and abitrary consumption amongst the populace who have become adicted to it. This research was designed to investigate its effects on serum liver and kidney function parameters in *Sprague Dawley* rats. This is to ascertain hepatic and renal function or dysfunction potentials of a seed becoming indiscriminately ingested.

2. MATERIALS AND METHODS

Plant material: *Garcinia kola* were obtained from a local Market in Ado-Ekiti and certified at the herbarium unit of the Department of Botany, Ambrose Alli University.

Extraction of *Garcinia kola:* The *Garcinia kola* was rinsed in clean water to remove any sand, dirt and contaminant and then cut into small pieces. The pieces of the *Garcinia kola* was dried at 45°C for 72 hours using a hot air convection oven (Model SM-9023), milled with Glen Creston Laboratory mill (Model HA7 1DA) and then passed through a 40 mesh sieve. The powdered sample was soaked in distilled water (20% W/V) for 48 hours (2 days) with intermittent stirring/shaking. At the end of the extraction, the extract was filtered using muslin cloth to remove all matters that are not extractable, including cellular materials and other constituents that are insoluble in the extraction solvents. Afterward, the filtrate was concentrated on a water bath set at 45°C.

Animal selection: The animals were obtained from the Animal House of the College of Medicine, Ambrose Alli University. Ekpoma. A total of 18 adult female rats weighing between 150 to 200g were used. The rats were kept in cages in the animal house and were allowed acclimatization for 2 weeks before the commencement of the experiment. Room temperature was between 30-31% while room barometric pressure was 750-760mmHg. The rats were fed *ad libitum* with grower's mash and had free access to water. The rats were weighed for the period of the experiment.

The 18 female adults rats were divided into 3 groups as follows: Group I serves as the control, Group II and III received *Garcinia kola* extract at doses of 0.5 mg/kg or 1 mg/kg orally for 2 weeks. Administration was done orally through the means of a metal cannula as described by Naiho and Ugwu with slight modification [16]. The animals were weighed weekly.

Sample collection: The animals were sacrified by sedating with chloroform. Blood samples were collected from all the rats through cardiac puncture into tubes without coagulant and allowed to clot. The blood was centrifuged afterward at 4,000 rpm for 10 minutes to obtain the serum for determination of total bilirubin concentration (using the method described by Tietz and WHO) and for the assay of the activities of alkaline phosphatase (ALP) (using assay kit from Quimica Clinica, Spain) and alanine transaminase (ALT) and aspartate transaminase (AST) (Using an assay kits from Randox Laboratories, UK) [17, 18]. The serum electrolytes, urea and creatinine were determined by colorimetric method using commercial kits.

Statistical analysis: The Statistical Package for Social Sciences (SPSS version 20) was used for data analysis. The one-way analysis of variance (ANOVA) was employed for data analysis and where applicable LSD was determined and confidential interval of $p \leq 0.05$ considered statistically significant. Results were presented as mean ± Standard error of mean (SEM) using suitable tables and charts.

3. RESULTS

Effect of *Garcinia kola* on liver function parameters: As shown in Table 1, there was a significant increase in the serum ALT in the group treated with 1mg/kg of *Garcinia kola* extract compared with control and group II.

However there was no significant difference in the serum AST, bilirubin, albumin and total protein across the groups.

Effect of *Garcinia kola* on kidney function parameters: Table 2, shows significant increase in serum potassium, urea and creatine in the group treated with 1mg/kg body weight of *Garcinia kola* extract compared with control and group II. However, there was no significant difference in the serum level of sodium, chloride and bicarbonate across the the groups.

Table 1: Effect of *Garcinia kola* on liver function parameters

	Group I (control)	Group II (0.5mg/kg)	Group III (1mg/kg)
ALT (u/l)	18.60±1.96	16.00± 1.79	26.00±1.41^{*+}
AST (u/l)	16.80±1.88	16.80±2.42	19.80±0.74
ALP (u/l)	48.00±2.61	38.20±5.56	57.20±6.35^{+}
Bilirubin (g/dl)	0.20±0.18	0.21±0.25	0.22±0.01
Albumin (g/dl)	51.80±6.74	50.00±2.68	47.20±2.52
Total Protein (g/dl)	74.60±4.60	69.40±3.57	79.20±6.34

Values are means ± SEM. * indicates significant difference at $p<0.05$ compared with control (group A); + indicates significant difference at $p<0.05$ compared with Group II

Table 2: Effect of *Garcinia kola* on kidney function parameters

	Group I (control)	Group II (0.5mg/kg)	Group III (1mg/kg)
Potassium (mmol/l)	4.16±0.25	4.62±0.11	5.58±0.32*$^{+}$
Sodium (mmol/l)	137.20±2.52	138.00±0.95	137.40±0.81
Chloride (mmol/l)	98.40±0.51	99.40±0.40	99.20±0.37
HCO_3^- (mmol/l)	22.80±0.58	24.60±0.93	24.40±0.40
Urea (g/dl)	27.60±12.34	41.20±13.67	110.00±14.65^{*+}
Creatinine (g/dl)	0.72±0.04	0.88±0.37	1.76±0.05^{*+}

Values are means ± SEM. * indicates significant difference at $p<0.05$ compared with control (group A); + indicates significant difference at $p<0.05$ compared with group II.

4. DISCUSSION

Certain phytochemicals that could be found in the *Garcinia kola* (oxalates and cardiac glycosides) have been reported to be toxic to vital organs and tissues [19, 20]. When a vital organ such as the liver is damaged by chemical agents, loss of organ function may be determined by certain biochemical markers such as serum total bilirubin concentration and the activities of serum ALP, ALT and AST. AST and ALT are often released into the blood stream when liver cells are destroyed by a chemical agent, and this release is often in conformity with the extent of the damage of the liver cells [21]. Elevated serum ALP activity and bilirubin concentration are indicative of bile ducts obstruction. Bilirubin may also be elevated when the secretory function of the liver is impaired. These biochemical makers are useful for screening for liver disease, monitoring of side effects of medications, and in determining responses to treatment for a given liver disease [22].

The result of this study shows significant increased serum ALT and ALP in the group treated with 1mg/kg body weight of *Garcinia kola* extract, while the the group treated with 0.5mg/kg body weight of the extract does not appear to alter the serum level of these biomakers. In line with the understanding of the serum elevation of these biomakers in relation to hepatic damage, heavy consumption

of *Garcinia kola* seed may be harmful to the liver health and should be discouraged. According to Kim et al [23], a mild increase in liver enzymes correlates with an increased risk of future mortality. The finding of this study is in line with that of Anyia et al [24], that *Garcinia kola* significantly increased plasma levels of total bilirubin, conjugated bilirubin, AST and ALT but contrary to the findings of Emine et al., who reported that biomarkers of liver function such as serum total bilirubin concentration, and activities of serum AST, ALT and ALP were all found to be unaffected by the kola seed feeding [25].

Regular kidney function tests are essential to ensure safe monitoring and optimal treatment of longterm health conditions such as heart failure, diabetes mellitus, and chronic kidney disease [26]. Kidney helps in maintaining homeostasis of the body by reabsorbing important material and excreting waste products [27]. Its function is usually assessed by the levels of electrolytes, urea and creatinine in the blood. Renal diseases which diminish the glomerular filteration lead to urea retention and decrease in urea is seen in severe liver disease with destruction of cells leadings to impairment of the urea cycle [28]. Creatinine retention in the blood is evidence of kidney impairment. The result of this study also showed a significant stimulation of blood potassium, urea and creatinine at higher dose of *Garcinia kola* extract. The reported medicinal effects of *Garcinia kola* seed may have underlying involvement of alterations in body levels of trace metals and electrolytes as previously documented [29].

CONCLUSION

This study showed *Garcinia kola* extract has significant increasing effect on serum ALT, urea and creatinine at a higher dose and therefore should be consumed with moderation.

REFERENCES

1. Nelisiwe P. M., Jeremiah O. U., Sogolo L.L. Promising role of medicinal plants in the regulation and management of male erectile dysfunction. *Biomed Pharmacother*, 2020; 130; 110555.
2. Akoachere J.F., Ndip R.N., Chenwi E.B., Ndip L.M., Njock T.E., Anong D.N. Antibacterial effect of Zingiber officinale and Garcinia kola on respiratory tract pathogens. *East Afr Med J.* 2002; 79(11): 588-592.
3. Eminedoki D.G., Uwakwe A.A. and Ibe, G.O. Protective effect of Garcinia kola seed and honey mixture against paracetamol-induced hepatotoxicity in rats. *Nig. J. Biochem. Mol. Biol*. 2010; 25(2):1-8
4. Udenze, E.C.C, Braide, V.B, Okwesilieze, C.N. and Akuodor, G.C. Pharmacological effects ofGarcinia kola seed powder on blood sugar, lipid profile and atherogenic index of alloxan-induced diabetes in rats. *Pharmacologia*, 2012; 3 (12); 693-699.
5. Ugboko HU, Nwinyi OC, Oranusi SU, Fatoki TH, Omonhinmin CA. Antimicrobial Importance of Medicinal Plants in Nigeria. *Sci World J.* 2020; 22: 7059323.
6. Njume C., Afolayan A.J, Clarke A.M. And Ndip R.N. Crude ethanolic extracts of Garcinia kola seeds Heckel (Guttiferae) prolong the lag phase of Helicobacter pylori; Inhibitory and bactericidal potential. *J. Med. Food*, 2011; 14 (7-8); 822-827
7. Sun X, Yuan Y, Xin X, Sun P, Sun Y, Xie M, Wang Y, Huang Sh, Li B.

Ethanol extract of Garcinia kola seeds alleviates HGHFD/STZ-induced nonalcoholic fatty liver disease in diabetic rats by modulating oxidative stress, inflammation, and lipid accumulation. *Iran J Basic Med Sci;* 2025; 28: 662-670.

8. Buba CI, Okhale SE and Muazzam I. Garcinia Kola: The Phytochemistry, Pharmacology and Therapeutic Applications. *International Journal of Pharmacognosy,* 2016; 3(2): 67-81.
9. Udia P.M., Braide V.B., Owu D.U. Antispasmodic and spasmolytic effects of methalonic extract from seeds of Garcinia kola on isolated rat small intestine. *Niger J Physiol Sci.* 2009; 24(2) 111-6
10. Oluwole F.S. and Obafemi A.B. The effect of Garcinia kola on Gastric acid secretion in Albino rats *Nig. J. of Physiol Sci.* 1992; 8(1 – 2) p 115.
11. Elekwa I. Monanu M.O. Anosike E.O. Effect of aqueous extract of Garcinia kola Seed on membrane stability of Hb AA, HbAS and HbSS human erythrocytes. *Global J Med Sci*., 2003; 2(2): 97-101.
12. Naiho A.O. Effect of Gacinia kola on reproductive organ weight and histology of Testis and anterior pituitary gland of wistar rats. *J. Exp. and Clin. Anatomy*. 2004; 3(2): 20-23.
13. Braide V. B. Pharmacological effects of chronic ingestion of Garcinia kola seeds in rats. *Phytother. Res.* 1990; 4: 39-41.
14. Braide V. B. and Grill, V, Histological alterations by a diet containing seeds of Garcinia kola: effect on liver, kidney intestine in the rat. *Gegenbaurs Morphol. Jahrb*.,1990; 136(1): 95-101.
15. Adegboye M.F., Akinpelu D.A. and Okoh A.I. The bioactive and phytochemical properties of Garcinia kola (Heckel) seed extract on some pathogens. *Afr. J. Biotechnology*, 2008; 7:3934-3938.
16. Naiho A.O. and Ugwu A.C. Blood Pressure Reducing Effect of Bitter Kola (Garcinia kola, Heckel) in Wistar Rats. *Afr. J. Biomed. Res.*2009; 12(2): 131-134.
17. Tietz, N.M. Fundamentals of clinical Chemistry. W.B. Saunders Company, 1986; pp 1388-1390.
18. WHO (World Health Organization). Blood safety and clinical technology Guidelines on standard operating procedures for clinical Chemistry. Website: www.saero.who.int/en section 10/ section 17/ section 53/section 4. (Accessed on6/8/2010.
19. Monago C.C and Akidue V. Estimation of tannin, saponin, oxalate, cyanogenic and cardiac glycosides in Garsinia kola.*Sci Environ. Mgmt.* 2002; 6 (I) 22-25.
20. EMEA (European Medicines Agency). Committee for veterinary medicinal products. Oxalic acid. Summary report. EMEA/MRL/891/03. FINAL, December, 2003.
21. Lin J.K. and Wang C.J. Protection of crocin dyes in the acute hepatic damage induced by aflatoxin B1 and dimethyInitrosamine in rats. *Carcinogenesis*, 1986; 7:595-599.
22. Lee T.H., Kim W.R., Poterucha J.J. Evaluation of Elevated Liver Enzymes. *Clin Liver Dis*., 2012; 16: 183–198.
23. Kim H.C., Nam C.M., Jee S.H., et al. Normal serum aminotransferase concentration and risk of mortality

from liver diseases: prospective cohort study. *BMJ*; 2004; 328: 983.

24. Anyia FA, Bamigbowu OE, Obisike UA. Effect of dietary consumption of Garcinia kola seed and horney mixture on liver function parameters of wista rats. *International Research Journal of Gastroenterology and Hepatology* 2025; 8(1) , 123-132.
25. Tamuno-Emine DG, Ben-Chioma AE, Uwakwe AA. Effect of Garcinia kola seed on serum lipid profile and enzyme activities of healthy human subjects. *Journal of Biology and Genetic Research*, 2015; 1(8): 17-22
26. Murray J.S., Williams C.J., Lendrem C., Smithson J., Allinson C., Robinson J., Walker A., Winter A., Simpson A.J., Newton J., Wroe C. and Jones W.S. Patient Self-Testing of Kidney Function at Home, a Prospective Clinical Feasibility Study in Kidney Transplant Recipients. *Kidney Int Rep*. 2023; 8, 1170–1182
27. Judd E, Kumar V, Porrett PM, Hyndman KA, Anderson DJ, Jones-Carr ME, Shunk A, Epstein DR, Fatima H, Katsurada A, Satou R, Navar LG, Locke JE. Physiologic homeostasis after pig-to-human kidney xenotransplantation. *Kidney Int*. 2024; 105(5):971-979.
28. Ranjna C. Practical clinical biochemistry methods and interpretation. 2nd ed. Delhi: Jaypee Brother Medical Publishers (P) Limited, 1999; p. 117.
29. Agada P. O. and Braide V. B. Effect of dietary garcinia kola seed on selected serum electrolytes and trace metals in male albino rats. *Nigerian Journal of Physiological Sciences*, 2009; 24 (1-2): 53 – 57.